

Tactile Robot Positioning & Calibration Guide

This guide summarizes the chain of logic for calculating the pose of your tactile sensor pads and how to implement this in your system.

1. The Positioning Chain

To find where your tactile pads are in space:

- **Start with FK:** Use Forward Kinematics to find the exact position/orientation of the End-Effector (EE) base.
- **Apply Offsets:** Add the fixed static transformation from the EE base to the fingertip base.
- **Incorporate Finger State:** Add the dynamic offset (based on current open/close width) to find the fingertip position.
- **Final Pad Transform:** Apply the fixed transformation from the fingertips to the center of the tactile pads.

2. Necessary Data Inputs

To make this chain work, you must define and store:

- **Fixed Transforms:** The rigid distance/rotation from the EE base to the finger base, and from the finger base to the tactile pad surface.
- **Finger Range:** The distance of the fingertips from the base at "Full Open" vs. "Full Closed" positions.
- **Object Feedback:** The actual distance the gripper is currently closed (blocked width), which determines your current Z-offset for the fingers.

3. Improving Your GUI Calibration Tab

To improve calibration, your GUI should allow you to:

- **Measure & Input:** Manually define or automatically record the "Full Open" and "Full Closed" values.
- **Live Feedback:** Display the current "blocked" width of the gripper when an object is held.
- **Offset Calculator:** A utility that updates the finger-offset variable in real-time based on the object width, ensuring the kinematic chain is always current.

4. Commanding Grid Movements

To move the tactile pad to a specific target in the workspace:

- **Inverse Mapping:** Calculate the required EE pose by working backward from your desired pad location (subtracting the pad and finger offsets).
- **IK Execution:** Pass this calculated EE pose into your Inverse Kinematics (IK) solver to generate the required joint movements for the robot.

- **Grid Logic:** Treat your grid as a sequence of target EE poses, ensuring each pose adjustment accounts for the current object width (the current finger offset).